





# Pinguecula eyes' problem that does matter.

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**What makes you  
Pinguecula'ed ?**



**The moment you  
leave an indoor room  
matters.**

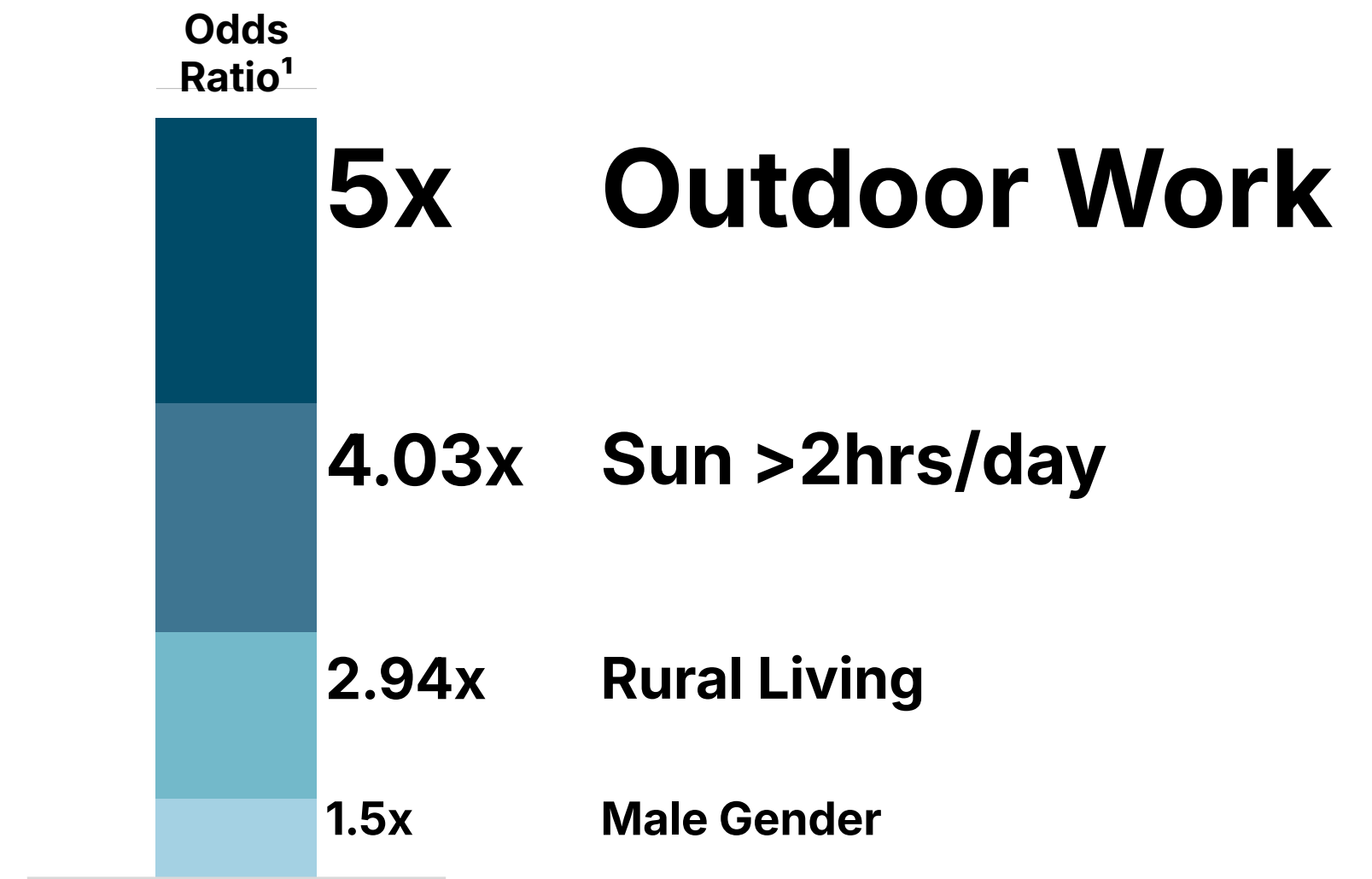


# Statistical Risk Factors: UV & Occupation

## Correlation Significances

- Direct sun exposure, work type, cumulative sunlight, living area, gender, age, education are significant causes of Pinguacula.
- Every unit of cumulative sunlight increases risk by 2.1x (Taiwan Study) and peaks at 60-69 years old.

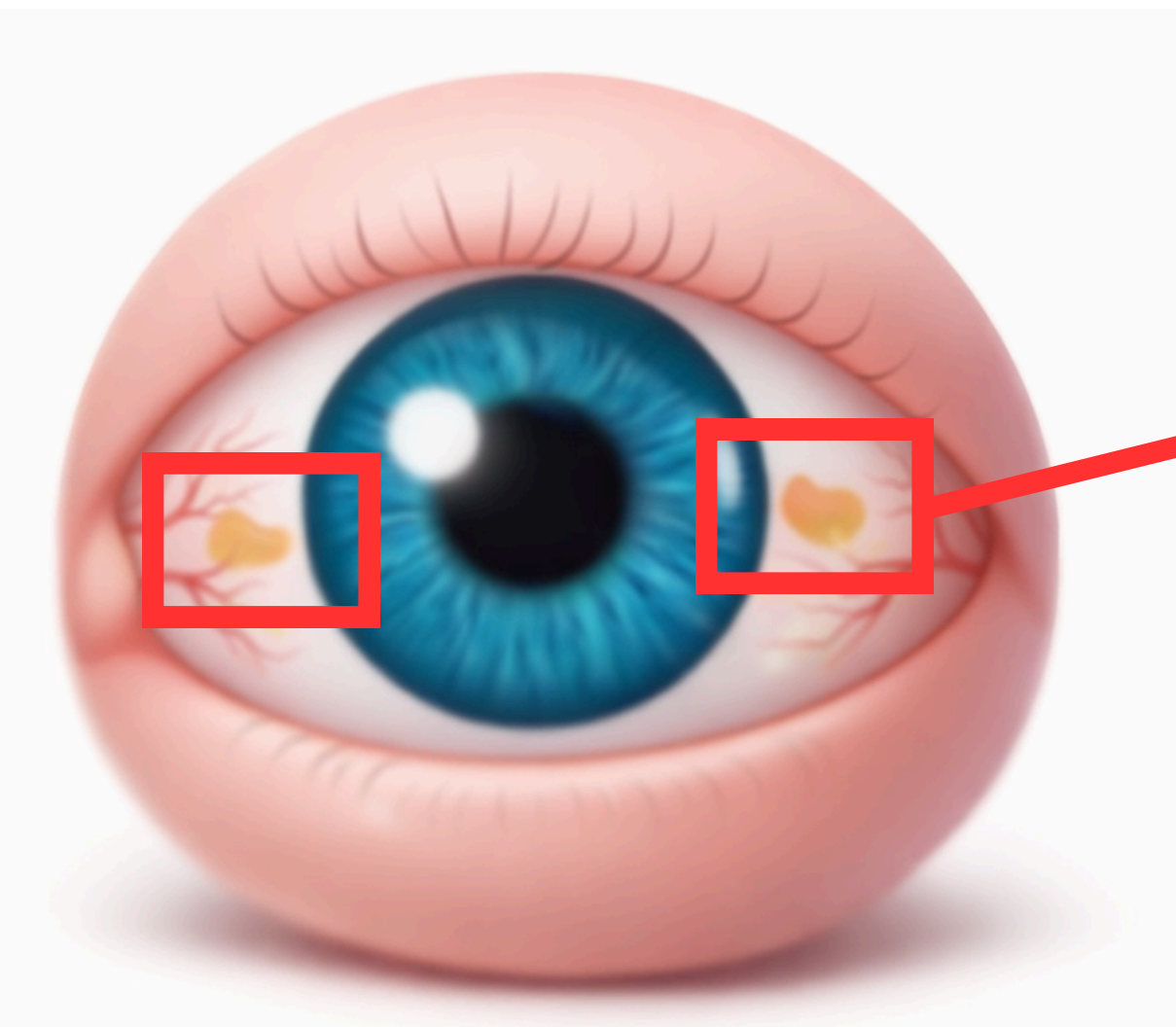
## Relative Risk of Pinguacula



<sup>1</sup> Odds ratio is risk multiplier of association between an exposure and an outcome.

Data sources: Studies from Western Turkey, Taiwan, and Australia

# 3 Things about Pinguecula



- Common degeneration
- Forms on nasal side or temporal side of white eye
- Redness, itching, tearing sometimes

# Reducing the Odds: Evidence-Based Prevention

## Protective Factors<sup>2</sup> and Clinical Prevention

### Explicit Statistical Protective Factors (OR<1)



#### Indoor Work

**80% less likely to develop Pterygia** compared to outdoor workers.



#### Higher Education

individuals have a **43% lower risk**.

### Practical Protective Behaviors



#### 100% UV Block

is **primary physical** barrier against the UV radiation



#### Brimmed Hats

reduces the amount of indirect sunlight reaching the eye from above



#### Moisture Control

avoids wind, dust, and smoke (co-factors)

<sup>2</sup> OR < 1, the factor reduces risk.



# Preventing the problem can begin now.

100% UV-blocking  
sunglasses and  
brimmed hats



Most Common

Artificial tears  
for dryness and  
mild steroid  
drops for  
inflammation



Suppliment

Surgical Excision  
with Graft

\$\$\$



Removing the  
bump and  
patching the area  
  
~45,000 THB as  
available stand  
offering

Gold Standard Cure

PANIS (Plasma-  
Assisted Noninvasive  
Surgery)

?????



Using PLEXR  
device and  
vaporizing the  
bump without  
cutting or stitches  
  
Unavailable  
currently in the  
Eye center.  
Patients are  
advised to look for  
specialised clinics.

Future Innovation

# Wrap-Up: What to Be Aware Of

## Problem

Eyes' **common degeneration** forms on nasal side or temporal side of white eye that sometimes **makes redness, itching and tearing**.

## Cause

Studies show that **environment, behavior, demographic factors** multiply the risk of Pinguecula.

## Solution

Always **keep your eyes moisturized** and **avoid direct sun exposure**. The standard treatment is **surgical excision**, with an estimated cost of **45,000 THB**.

A potential **future treatment involves plasma vaporization**, but it has not yet been commercialized or proven in practice.



**The priority is yours—protect your vision and warn your loved ones, because once this disease develops, your life changes forever.**

**With all cares,  
Group 9 team :)**

**Thank you for attention,  
Feel Free to discuss!**

# Reference

## Primary Clinical Source

- Bangkok Hospital (2025). Health for Life: Pinguecula and Pterygium – Degeneration of the Conjunctiva. Medical Review by Dr. Tarinee Sangiampornpanit (Ophthalmology).  
<https://www.bangkokhospital.com/en/bangkok/content/pinguecula-and-ptyerygium-degeneration-of-the-conjunctiva>.

## Statistical Epidemiology

- Dündar, H. et al. (2020). Prevalence and associated factors of pinguecula in western Turkey. Pamukkale Medical Journal. (Source for: Age peaking at 60-69 and gender risks).
- Panchapakesan, J. et al. (1998). Prevalence of pterygium and pinguecula: the Blue Mountains Eye Study. Australian and New Zealand Journal of Ophthalmology. (Source for: 69.5% prevalence in older populations).
- Asokan, R. et al. (2012). Prevalence and associated factors for pterygium and pinguecula in a South Indian population. Ophthalmic and Physiological Optics. (Source for: Rural vs. Urban odds ratios).

## Occupational Risk Factors

- Tang, F. C. et al. (1999). Relationship between pterygium/pinguecula and sunlight exposure among postmen in central Taiwan. Chinese Medical Journal. (Source for: 2.1% risk increase per unit of sunlight).
- Ay, İ. et al. (2022). Ocular health among industrial workers. La Medicina del Lavoro. (Source for: Industrial exposure risks beyond sunlight).
- Tesfai, B. et al. (2021). Prevalence of Solar Keratopathy, Pterygium... in the Islands of Northern Red Sea Zone, Eritrea. Clinical Ophthalmology. (Source for: High risk in fishermen).

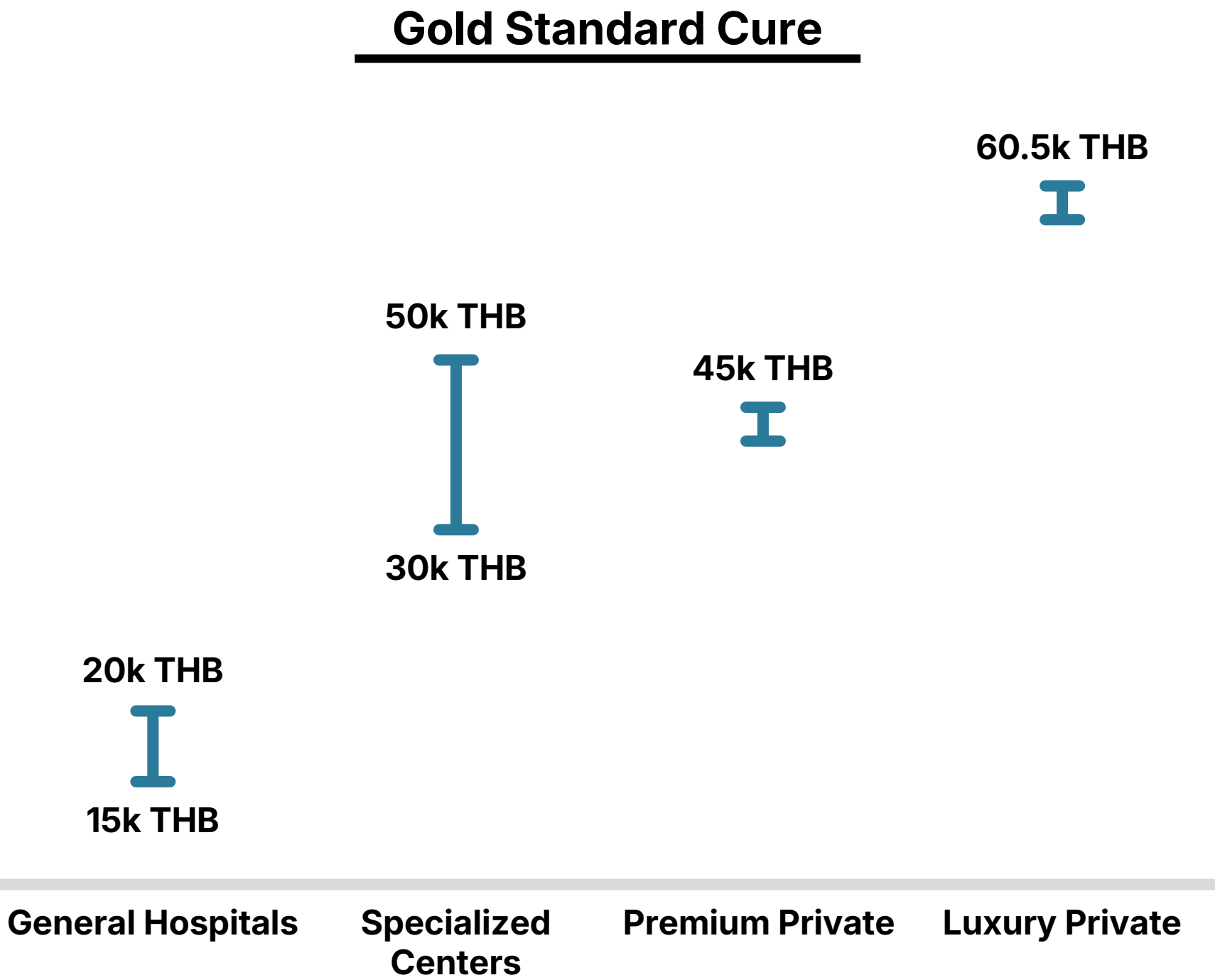
## Environmental Mechanisms

- Hatsusaka, N. et al. (2021). Association among pterygium, cataracts, and cumulative ocular ultraviolet exposure. PLoS ONE. (Source for: Cumulative UV dose-response).
- Zeng, H. et al. (2025). The nonlinear impact of air pollutants and solar radiation exposure on the risk of hospitalisation for pterygium. Journal of Global Health. (Source for: Emerging link to air pollution).
- Fekadu, S. et al. (2020). Prevalence of pterygium and its associated factors... in Southwest Ethiopia. PLoS ONE. (Source for: Efficacy of sunglasses/hats).



# Appendix:

## Clarification for clinical treatment prices



Examples of Thai eye clinics and specialized centers

- |                            |                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| <b>General Hospitals</b>   | <ul style="list-style-type: none"><li>• Ladprao Hospital</li><li>• Camilian</li></ul> |
| <b>Specialized Centers</b> | <ul style="list-style-type: none"><li>• Rutnin Eye Hospital</li></ul>                 |
| <b>Premium Private</b>     | <ul style="list-style-type: none"><li>• Bangkok Hospital</li></ul>                    |
| <b>Luxury Private</b>      | <ul style="list-style-type: none"><li>• Bumrungrad Hospital</li></ul>                 |